Mission 1.1: Power Station — Dialogue Script

Location: Power Station & Habitat Grid
Characters: Chiaki Mukai (Astronaut / Power Engineer), Astro Bee (AI Assistant), Neil Armstrong (Astronaut)

Scene 2 — Station Welcome

- **Speaker**: Chiaki Mukai
- **Dialogue**:
  "Welcome to the power station! Here we generate, store, and route electricity."

Scene 3 — Photovoltaic Generation

- **Speaker**: Chiaki Mukai
- **Dialogue**:
  "This is our Solar Array![**1**] Photovoltaic cells convert sunlight into electrical power.[**2**]"

[1] What Powers a Spacecraft? | NASA Space Place – NASA Science for Kids

https://www.nasa.gov/learning-resources/nasas-picture-dictionary-s/

[2] 3.0 Power - NASA

Scene 4 — Polar Siting

- **Speaker**: Chiaki Mukai
- **Dialogue**:
  "Tall deployable arrays[**3**] at the south pole capture continuous sunlight above craters.[**4**]"

[3] NASA, Industry to Mature Vertical Solar Array Technologies for Lunar Surface

Vertical Solar Array Technology (VSAT)

[4] Shadows near the Moon's South Pole - NASA Science

Shackleton Crater's Illuminated Rim & Shadowed Interior - NASA Science

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Scene 5 — Energy Storage

- **Speaker**: Chiaki Mukai
- **Dialogue**:
  "This Battery Unit stores energy for use when solar arrays cannot produce power.[**5**]"

[5] Integrated Lunar Power Strategy Considerations

Lunar Surface Technology

https://ntrs.nasa.gov/citations/20220017550

Scene 6 — Lunar Day/Night Cycle

- **Speaker**: Chiaki Mukai
- **Dialogue**:
  "Lunar daylight lasts 14 Earth days, followed by 14 days of night[**6**]. Storage is vital [**7**]because..."

[6] https://ares.jsc.nasa.gov/interaction/lmdp/documents/TotheMoon_back.pdf

[7] https://www.lpi.usra.edu/lunar/strategies/resources/M2M%20ADD_ESDMD-001_NASA-TP-20230002706.pdf

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Scene 7 — Nightfall Power Conservation

- **Speaker**: Chiaki Mukai
- **Dialogue**:
  "Panels need sunlight and battery capacity is limited,[**8**] so we must conserve energy."

[8] Lunar Surface Technology

Scene 8 — Nuclear Surface Power

- **Speaker**: Chiaki Mukai
- **Dialogue**:
  "Nuclear fission systems[**9**] can generate continuous power even during long lunar darkness."

[9] Fission Surface Power - NASA

Scene 9 — Future Reactor Deployment

- **Speaker**: Chiaki Mukai
- **Dialogue**:
  "Fission reactors will power the outpost through darkness,[**10**] but none operate here yet.[**11**]"

[10] NASA's Fission Surface Power Project Energizes Lunar Exploration

[11] Fission Surface Power - NASA

Scene 10 — Grid Distribution

- **Speaker**: Chiaki Mukai
- **Dialogue**:
  "This Power Distribution Unit[**12**] routes electricity to life support and science labs."

[12] Electric Power on the Moon

Power

Scene 11 — Tour Conclusion

- **Speaker**: Chiaki Mukai
- **Dialogue**:
  "Great job! Next, we verify your knowledge before solar restoration."

Scene 12 — Astro Bee Evaluation Intro

- **Speaker**: Astro Bee
- **Dialogue**:
  "Let's do a quick system check to verify what you've learned, Cadet!"

Scene 13 — Knowledge Verification Q1

- **Speaker**: Astro Bee

- **Dialogue / Prompt**:
  "Q1. Which component generates electricity from sunlight?"[**13**]

[13]

Powering the Moon Using Vertical Solar Array Technology

Power and Energy for the Lunar Surface

    - A) Battery Storage Unit
    - B) Solar Array *(Correct)*
    - C) Power Distribution Unit

Scene 14 — Knowledge Verification Q2

- **Speaker**: Astro Bee
- **Dialogue / Prompt**:
  "Q2. What is the primary function of the Battery Storage Unit?"[]
    - A) Store energy for later use *(Correct)*
    - B) Transmit signals to Earth
    - C) Remove dust from solar panels

Scene 15 — Knowledge Verification Q3

- **Speaker**: Astro Bee
- **Dialogue / Prompt**:
  "Q3. What does the PDU do?"
    - A) Generates sunlight
    - B) Stores all the outpost's energy
    - C) Routes power to connected thkle equipment *(Correct)*

Scene 16 — Knowledge Verification Q4

- **Speaker**: Astro Bee
- **Dialogue / Prompt**:
  "Q4. What is a key potential advantage of fission surface power?"
    - A) It can provide power independently of sunlight. *(Correct)*
    - B) It requires sunlight to operate.

- C) It eliminates the need for power distribution.

Scene 17 — Knowledge Verification Q5

- **Speaker**: Astro Bee
- **Dialogue / Prompt**:
  "Q5. Why is operating continuously at the battery's maximum discharge rating a concern[**14**]?"

[14] Ames Technical Standard

3.0 Power - NASA

Guidelines on Lithium-ion Battery Use in Space Applications

- A) It leaves little or no discharge margin. *(Correct)*
- B) It automatically recharges the battery.
- C) It guarantees unlimited operating time.

Mission Complete

- **Speaker**: Astro Bee
- **Dialogue**:
  "Mission accomplished! Power station and habitat [**15**]are safe and fully operational!"

[15] Powering the Moon Using Vertical Solar Array Technology

Mission 2.1: Daily Meal & Food Storage — Dialogue Script

Location: Lunar Food Storage & Crew Dining Hall
Characters: Sunita Williams (Astronaut / Mission Specialist), Astro Bee (AI Assistant)

Scene 1 — Food Storage Welcome

- **Speaker**: Sunita Williams
- **Dialogue**:
  "Welcome to food storage[**1**], Cadet! Today we plan daily nutrition for our crew."

[**1**] 7.0 Habitability Functions - NASA

Scene 2 — Multilayer Pouch Technology

- **Speaker**: Sunita Williams
- **Dialogue**:
  "Meals are sealed in multilayer PET, aluminum, and PE film to block oxygen and moisture.[**2**]"

[2] Space Food Packaging Facts

https://www.nasa.gov/wp-content/uploads/2023/03/food-and-nutrition-technical-brief-ochmo.pdf

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Scene 3 — Preservation Atmosphere

- **Speaker**: Sunita Williams
- **Dialogue**:
  "Nitrogen flushing [**3**]keeps oxygen below 0.5% while climate control protects food quality."

[3] Space Food Packaging Facts

https://www.nasa.gov/wp-content/uploads/2023/03/food-and-nutrition-technical-brief-ochmo.pdf

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Scene 4 — Menu Selection

- **Speaker**: Sunita Williams
- **Dialogue**:
  "Choose your meal for today. Pick one of the three options below!"

Scene 5 — Kitchen Preparation

- **Speaker**: Sunita Williams
- **Dialogue**:
  "Great! Let's get your meal prepared."

Scene 6 — Food Categories

- **Speaker**: Sunita Williams
- **Dialogue**:
  "Space foods come in three types: Ready-to-Eat, Rehydratable, and Food Warmerable." [**4**]

[4] Space Food | NASA

Food in Space - NASA

Scene 7 — Ready-to-Eat Items

- **Speaker**: Sunita Williams
- **Dialogue**:
  "For ready-to-eat meals, simply tear the pouch open and enjoy!"

Scene 8 — Rehydratable Meals

- **Speaker**: Sunita Williams
- **Dialogue**:
  "For rehydratables, connect the pouch to the dispenser and inject potable water [**5**]."

[5] Crew Systems - NASA

NASA Space Food Formulations: Rehydratable Foods

Scene 9 — Electric Food Warmer

- **Speaker**: Sunita Williams
- **Dialogue**:
  "For hot meals, place the sealed pouch in the electric food warmer."

Scene 10 — Dining Hall Gathering

- **Speaker**: Sunita Williams
- **Dialogue**:
  "Your meal is ready! Let's eat together, Cadet."

Scene 11 — Nutrition Logging

- **Speaker**: Sunita Williams
- **Dialogue**:
  "Daily intake logged! Energy reserves balanced for lunar duty. [**6**]"

[6] 7.0 Habitability Functions - NASA *(Alternative Dynamic Text: "[chosenKcal] kcal logged! Energy balanced. Click SKIP for verification!")*

Scene 12 — Astro Bee Evaluation Intro

- **Speaker**: Astro Bee
- **Dialogue**:
  "Let's do a quick verification of what you've learned, Cadet!"

Knowledge Verification Q1

- **Speaker**: Astro Bee
- **Dialogue / Prompt**:
  "Q1. What is the average daily calorie requirement for an astronaut?"
    - A) 2,500 kcal
    - B) 3,035 kcal *(Correct)*
    - C) 3,500 kcal

Knowledge Verification Q2

- **Speaker**: Astro Bee
- **Dialogue / Prompt**:
  "Q2. Why is nitrogen flushing used in space food packaging?"[**7**]

[7] https://www.nasa.gov/wp-content/uploads/2023/03/food-and-nutrition-technical-brief-ochmo.pdf

    - A) To reduce oxygen and preserve food quality *(Correct)*
    - B) To increase food temperature
    - C) To add moisture to food

Knowledge Verification Q3

- **Speaker**: Astro Bee
- **Dialogue / Prompt**:
  "Q3. How do astronauts prepare rehydratable food?"
    - A) Eat it directly
    - B) Place it in a food warmer
    - C) Add the required amount of potable water *(Correct)*

Knowledge Verification Q4

- **Speaker**: Astro Bee
- **Dialogue / Prompt**:
  "Q4. How should astronauts warm food that requires heating?"
    - A) Place the sealed package in an electric food warmer *(Correct)*
    - B) Add extra water
    - C) Leave it at room temperature

Knowledge Verification Q5

- **Speaker**: Astro Bee
- **Dialogue / Prompt**:
  "Q5. What resources are consumed after selecting a meal in the game?"
    - A) 100 mL water and 15 Wh power

    - B) 150 mL water and 25 Wh power *(Correct)*
    - C) 200 mL water and 30 Wh power

Mission Complete

- **Speaker**: Astro Bee
- **Dialogue**:
  "Mission accomplished! Our lunar crew is well nourished and ready for duty!"

Mission 2.2: Hydroponics & Lunar Plant Growth — Dialogue Script

Location: Lunar Greenhouse & Hydroponics Plant Lab
Characters: Sunita Williams (Astronaut / Mission Specialist), Astro Bee (AI Assistant)

Scene 1 — Welcome & Experiment Kickoff

- **Speaker**: Sunita Williams
- **Dialogue**:
  "Welcome back, Rookie! Your first lunar soil experiment starts now."

Scene 2 — Soil Testing Purpose

- **Speaker**: Sunita Williams
- **Dialogue**:
  "Scientists have been exploring ways to grow food on the Moon. Today, you'll help us test a special growing system using treated lunar regolith and hydroponic nutrients."**[1]**

[1] Scientists Grow Plants in Lunar Soil - NASA

Moon Farming - NASA

Scene 3 — Regolith Collection Call

- **Speaker**: Sunita Williams
- **Dialogue**:
  "So let's collect some regolith."**[2]**

[2] What Is Lunar Regolith? The Moon's Surface Explained.

Scene 4 — Surface Airlock Hazard Warning

- **Speaker**: Sunita Williams
- **Dialogue**:
  "But first, why is lunar soil marked EXTREME DANGER?**[3]**
  [3] What Hazards Are Caused by Lunar Regolith? - NASA Science

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Scene 5 — Inhalation Danger

- **Speaker**: Sunita Williams
- **Dialogue**:
  "Its sharp glass particles can damage your lungs if inhaled,"[**4**]

[4] What Hazards Are Caused by Lunar Regolith? - NASA Science

What is Lunar Regolith? (Grades 5-8) - NASA

Scene 7 — Chemical Toxicity

- **Speaker**: Sunita Williams
- **Dialogue**:
  "While chromium compounds and reactive iron particles may pose health risks.[[5] Always use proper protection when handling regolith!"[**6**]

[5] Lunar Regolith Simulant User's Guide Revision A

Moon Farming - NASA

[6] The Challenge of Lunar Dust

Scene 8 — Lab Seeding & Safety

- **Speaker**: Sunita Williams
- **Dialogue**:
  "But don't worry, Rookie! You'll prepare the regolith safely,"

Scene 9 — Nutrients & Water Injection

- **Speaker**: Sunita Williams
- **Dialogue**:
  "add water and nutrients, and plant your seeds. Following the safety instructions carefully, let's see if we can grow food on the Moon!"

Scene 10 — Timelapse Prompt

- **Speaker**: Sunita Williams
- **Dialogue**:
  "Now press start for the timelapse!"

Scene 12 — Day 5 Growth Observation

- **Speaker**: Sunita Williams
- **Dialogue**:
  "Good work, Rookie! Compared with Earth-soil plants, lunar-regolith plants were smaller and showed signs of stress."[**7**]

[7] Scientists Grow Plants in Lunar Soil - NASA

Growing Plants in Lunar Soil - NASA Science

Scene 13 — Astro Bee Question 1 (Plant Size)

- **Speaker**: Astro Bee
- **Dialogue / Prompt**:
  "Q1. Observe the plants and select their size:"
    - Option 1: Normal
    - Option 2: Small *(Correct)*
    - Option 3: Large

Scene 14 — Day 15 Pigmentation Observation

- **Speaker**: Sunita Williams
- **Dialogue**:
  "Look closely! Some leaves are turning reddish. Earth-based experiments also found reddish pigmentation in plants grown in lunar regolith, possibly due to stress."[**8**]

  [8] Scientists Grow Plants in Lunar Soil - NASA

Scene 15 — Astro Bee Question 2 (Leaf Color)

- **Speaker**: Astro Bee
- **Dialogue / Prompt**:
  "Q2. Select the leaf color:"
    - Option 1: Green
    - Option 2: Slightly Red *(Correct)*
    - Option 3: Gray

Scene 16 — Day 23 Root & Substrate Stress

- **Speaker**: Sunita Williams
- **Dialogue**:
  "Lunar regolith lacks organic matter and has physical and chemical properties that can challenge plant roots and nutrient uptake, which stressed plants as observed in Earth-based experiments."

Scene 17 — Astro Bee Question 3 (Stress Level)

- **Speaker**: Astro Bee
- **Dialogue / Prompt**:
  "Q3. How stressed do the plants look?"[**9**]

  [9] NASA OSDR: OSD-476

  Scientists Grow Plants in Lunar Soil - NASA

    - Option 1: Low
    - Option 2: Moderate
    - Option 3: High *(Correct)*

Scene 18 — Apollo Mission Botanical Data (2022 Study)

- **Speaker**: Sunita Williams
- **Dialogue**:
  "In a 2022 study, scientists grew Arabidopsis plants in real lunar samples from Apollo missions. After 20 days, the plants were generally smaller and developed more slowly than controls grown in lunar soil simulant."

Scene 19 — Gene Activity Lab Findings

- **Speaker**: Sunita Williams
- **Dialogue**:
  "Their gene activity also revealed responses associated with salt, metals, and oxidative stress."

Scene 20 — Astro Bee Question 4 (Growth Rate)

- **Speaker**: Astro Bee
- **Dialogue / Prompt**:
  "Q4. How was the growth rate?"
  - Option 1: Slow *(Correct)*
  - Option 2: Moderate
  - Option 3: Fast

Scene 21 — Concluding Observation Report

- **Speaker**: Astro Bee
- **Dialogue**:
  "Thank you for your report! Your observations will help humanity learn more about lunar conditions."

Mission Complete

- **Speaker**: Astro Bee
- **Dialogue**:
  "Outstanding observations, Cadet! Your lunar regolith research brings us one step closer to sustainable agriculture on the Moon!"[**10**]

  [10] Scientists Grew Plants in Moon Soil — What Happened Next?

  Season 5, Episode 26: How to Grow Food on the Moon - NASA

Mission 5.1: Space Radiation Shelter — Dialogue Script

Location: Surface Perimeter & Radiation Storm Shelter
Characters: Rakesh Sharma (Astronaut / Safety Officer), Astro Bee (AI Assistant)

Scene 1 — Routine Surface Inspection

- **Speaker**: Rakesh Sharma
- **Dialogue**:
  "Let's check the greenhouse power before heading back."

Scene 2 — Space Radiation Alarm Trigger

- **Speaker**: Astro Bee
- **Dialogue**:
  "WARNING! Solar radiation level is rising!"[**1**]

[**1**] Design for Ionizing Radiation Protection OCHMO-TB-020 Rev G

Scene 3 — Solar Storm Evacuation Alert

- **Speaker**: Rakesh Sharma
- **Dialogue**:
  "A solar storm is approaching! Unlike Earth, the Moon has no natural protection against radiation.[**2**] Quick, let's get inside the radiation shelter!"

  [**2**] https://www.nasa.gov/wp-content/uploads/static/history/alsj/WOTM/WOTM-Radiation.html

  Small Steps, Giant Leaps: Episode 90, Effects of Space Radiation - NASA

Scene 4 — Space Radiation Telemetry Briefing

- **Speaker**: Rakesh Sharma

- **Dialogue**:
  "Space radiation consists of tiny, energetic particles[**3**] that can harm astronauts, even though we can't see them. There are two main types: Solar Particle Events (SPEs) and Galactic Cosmic Rays (GCRs)." [**4**]

  [**3**]
  https://ntrs.nasa.gov/api/citations/20210018122/downloads/Glenn%20Symposium%20Panel%20charts_2021%20-%20Jonathan%20Ransom.pdf

  [**4**] Protecting Astronauts from Space Radiation on the Lunar Surface

  Protecting Crew and Surface Systems with a Long-Duration Lunar Safe Haven

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Scene 5 — Solar Particle Events (SPEs)

- **Speaker**: Rakesh Sharma
- **Dialogue**:
  "SPEs come from the Sun and can cause sudden radiation spikes [**5**]during solar storms. That's why we must act quickly when a warning appears."

  [**5**] Design for Ionizing Radiation Protection OCHMO-TB-020 Rev G

Scene 6 — Galactic Cosmic Rays (GCRs)

- **Speaker**: Rakesh Sharma
- **Dialogue**:
  "GCRs come from beyond our Solar System.[**6**] They're always present and difficult to block, even when the Sun is calm."

  [**6**]
  https://ntrs.nasa.gov/api/citations/20210018122/downloads/Glenn%20Symposium%20Panel%20charts_2021%20-%20Jonathan%20Ransom.pdf

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Scene 7 — Earth vs. Moon Natural Shielding

- **Speaker**: Rakesh Sharma
- **Dialogue**:
  "Earth's magnetic field deflects many charged particles, and the atmosphere absorbs and deflects those particles. But the Moon lacks these natural shields, [**7**] so we must build our own protection."

[**7**] https://www.nasa.gov/wp-content/uploads/static/history/alsj/WOTM/WOTM-Radiation.html

Integration of Advanced Structures and Materials Technologies for a Robust Lunar Habitat

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Scene 8 — Radiation Shelter Entrance & Objective

- **Speaker**: Rakesh Sharma
- **Dialogue**:
  "Now that we're safe inside the shelter, let's protect our shelter from radiation! Placing materials between the radiation and the crew reduces exposure."[**8**]

  [**8**] Design for Ionizing Radiation Protection OCHMO-TB-020 Rev G

  Protecting Astronauts from Space Radiation on the Lunar Surface

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Scene 9 — Supplies Rearrangement Plan

- **Speaker**: Rakesh Sharma
- **Dialogue**:
  "We already have water containers, food, and other supplies[**9**] . Let's arrange them around the crew to strengthen our shield!

  [**9**] https://ntrs.nasa.gov/api/citations/20160006051/downloads/20160006051.pdf

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Scene 12 — Shield Construction Complete

- **Speaker**: Rakesh Sharma
- **Dialogue**:
  "Excellent work, Cadet! You've arranged our supplies to improve the shelter's radiation shielding. But we can do even more."

Scene 13 — Base Exterior Habitat Resource

- **Speaker**: Rakesh Sharma

- **Dialogue**:
  "Look outside—there's a resource all around us that could help protect our permanent habitat. Our crew isn't the only thing we need to protect."

Scene 14 — Lunar Regolith Shielding

- **Speaker**: Rakesh Sharma
- **Dialogue**:
  "The Moon is covered in a loose material called lunar regolith[**10**], which we can use to provide additional radiation shielding around and over parts of our habitat."[**11**]

  [**10**] What Is Lunar Regolith? The Moon's Surface Explained.

  [**11**] Lunar In-Situ Resource Utilization

Scene 15 — In-Situ Resource Utilization (ISRU) Engineering

- **Speaker**: Rakesh Sharma
- **Dialogue**:
  "Using regolith means we won't have to transport every kilogram of shielding material from Earth. Remember, smart engineering is about the best use of the resources available to us!"

Scene 16 — Mission Complete / Final Telemetry

- **Speaker**: Astro Bee
- **Dialogue**:
  "Thank you for your reports!! Your observations will help the humanities to learn more about lunar conditions."

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Now the ISS oxygen generation system and electrolysis.

Here is the ISRU script in your Power Station format. Underlined keywords carry a number, and each number points to a source on a NASA site (nasa.gov, science.nasa.gov, svs.gsfc.nasa.gov or ntrs.nasa.gov). Where a source is a video, the NASA page is the official one for that video.

Mission 3: ISRU Facility, Dialogue Script

Scene	Speaker	Dialogue	Source
Scene 1: South Pole Welcome	Guion Bluford	"Welcome to the lunar South Pole, Cadet! Over billions of years, water from comets, asteroids, and the solar wind may have accumulated on the Moon."	[1], [2]
Scene 2: Cold Traps	Guion Bluford	"Permanently shadowed craters act as natural 'cold traps' where sunlight never reaches the floor, allowing water ice to survive. Some of this ice may also be hidden beneath the surface!"	[1], [2], [3]
Scene 3: Start Exploring	Guion Bluford	"Let's explore the area and find where water ice might be present."	[4]
Scene 6: Promising Locations	Guion Bluford	"Great! We've identified some promising locations. Select one to choose our drilling site."	[3]
Scene 8: Trident Drill	Guion Bluford	"Excellent site, Cadet! Now, let's use the TRIDENT drill. It can drill through lunar regolith and bring samples from beneath the surface to the top, reaching depths of approximately one meter."	[5], [6], [8]
Scene 9: Sample Collected	Guion Bluford	"Now that we've collected our sample, it's time to analyze it."	[7]
Scene 10: MSOLO Spectrometer	Guion Bluford	"This is MSOLO, a mass spectrometer from NASA's PRIME-1 mission. It analyzes gases released from lunar soil to help identify water and other volatile substances."	[6], [7]

Scene	Speaker	Dialogue	Source
Scene 11: Analysis Findings	Astro Bee	"The analysis detected possible water-related substances! Further investigation will reveal how much water is present and if it can support future lunar missions."	[7], [9]
Scene 12: Collect Icy Soil	Guion Bluford	"Great work, cadet! We've found clues of water-related substances. Now, let's collect the icy lunar soil and recover the water trapped inside."	[9]
Scene 13: Robotic Scoop	Guion Bluford	"Our 'Robotic Lunar Regolith Scoop' collects ice-bearing regolith from the drilling area and transfers it into a sealed container for processing."	[10], [11]
Scene 14: Water Extraction Lab	Guion Bluford	"NASA has researched lunar water-extraction concepts that heat icy regolith and capture the released vapor. The condenser cools the vapor, turning it into liquid water."	[12], [13], [14]
Scene 15: Electrolysis Lab	Guion Bluford	"Now we will use the purified water from your previous mission."	[15]
Scene 16: Electrolysis of Water	Guion Bluford	"Transfer purified water into the electrolyser and activate it. It uses electricity to split water into hydrogen and oxygen."	[15], [16]
Scene 17: ISS Oxygen System	Guion Bluford	"The ISS uses an Oxygen Generation System (OGS) rack that houses the Oxygen Generation Assembly (OGA)."	[16], [17]
Scene 18: Oxygen Ready	Guion Bluford	"Oxygen is ready! Store it for the habitat's life-support system. The collected hydrogen can also be used as rocket propellant."	[9], [14], [15], [17]

References (NASA sources only)

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https://science.nasa.gov/solar-system/moon/nasas-lro-lunar-ice-deposits-are-widespread/

[2] 🎥 Water on the Moon, NASA Goddard (video)
https://svs.gsfc.nasa.gov/11756

[3] 🎥 The Moon's South Pole & Permanent Shadows, NASA Goddard Scientific Visualization Studio (videos)
https://svs.gsfc.nasa.gov/gallery/moonpole

[4] 🎥 Is There Water on the Moon? We Asked a NASA Scientist (Episode 27)

https://www.nasa.gov/general/is-there-water-on-the-moon-we-asked-a-nasa-scientist-episode-27

[5] Apollo to Artemis: Drilling on the Moon, NASA
https://www.nasa.gov/centers-and-facilities/kennedy/apollo-to-artemis-drilling-on-the-moon/

[6] PRIME-1 Science & Technology (TRIDENT and MSolo), NASA Science
https://science.nasa.gov/lunar-science/clps-deliveries/prime-1aiintuitive-machines/

[7] NASA's Lunar Drill Technology Passes Tests on the Moon, NASA

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[8] 🎥 PRIME-1 video, NASA
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[10] NASA Glenn Excavation Laboratory (robotic excavation of lunar regolith simulant), NASA
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[12] An Experimental Study on Low Pressure Frost Formation for Lunar Polar Water Capture, NASA Technical Reports Server
https://ntrs.nasa.gov/citations/20230010063

[13] Microwave Extraction of Water from Lunar Regolith Simulant, NASA Technical Reports Server
https://ntrs.nasa.gov/api/citations/20070018802/downloads/20070018802.pdf

[14] 🎥 How Will We Extract Water on the Moon? We Asked a NASA Technologist (Episode 47)

https://www.nasa.gov/general/how-will-we-extract-water-on-the-moon-we-asked-a-nasa-technologist-episode-47

[15] Environmental Control and Life Support Systems (ECLSS), NASA
https://www.nasa.gov/reference/environmental-control-and-life-support-systems-eclss

[16] Electrochemical Solutions for Advanced Life Support (ISS OGA electrolysis), NASA Technical Reports Server
https://ntrs.nasa.gov/citations/20200002346

[17] ISS Environmental Control and Life Support System fact sheet, NASA Marshall
https://www.nasa.gov/sites/default/files/atoms/files/g-281237_eclss_0.pdf

Notes

- **Underlining:** if your viewer shows the `<u>` tags as plain text, the underlined words are the ones right before each [number].
- **Videos:** I used NASA's own pages for videos instead of raw youtube.com links, because I could only confirm those pages. The Episode 27 and 47 pages link to NASA's YouTube playlist, and the Goddard pages (2 and 3) name the NASA Goddard YouTube channel.
- **How well I checked:** I matched sources to your lines using search results and page excerpts, not by reading every page in full. Please open each link once before shipping, especially [10], which is a short NASA link, and the technical papers [11], [12] and [13].
- **Scene 15:** "your previous mission" is not backed by any source, since it refers to your game's own flow.
- **Scene 14:** the 300°C heating shown in your lab simulation is a game number. The NASA papers support the heat-and-condense concept but not that exact figure.

I can also put this into a Word doc or a spreadsheet with columns for scene, speaker, dialogue and source if you'd like.